NYC Mobility Analytics Platform

Qeema Internship Project

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Date: November 2026

1. Project Overview

The objective of this project was to build a data analytics platform using New York City High-Volume For-Hire Vehicle (HVFHV) trip data.

The project involved extracting raw trip data, cleaning and validating it, loading it into PostgreSQL, designing a data warehouse, performing SQL analysis, optimizing queries, and creating an interactive dashboard using Apache Superset.

The final platform allows users to explore trip activity, identify patterns, compare providers, and analyze pickup locations and trip characteristics.

2. Tools and Technologies Used

The following tools were used throughout the project:

- **Python:** Data extraction and ETL processing.
- **Pandas:** Data manipulation and transformation.
- **PyArrow:** Reading Parquet files.
- **PostgreSQL:** Database storage and management.
- **SQL:** Data analysis and query optimization.
- **Apache Airflow:** Pipeline validation and database monitoring.
- **Apache Superset:** Dashboard creation and visualization.
- **Docker:** Running the Superset and Airflow environments.

3. Data Extraction and ETL Process

The project used NYC TLC High-Volume For-Hire Vehicle trip records covering January 2025 to August 2026.

A total of 20 monthly Parquet files were downloaded, with an approximate combined size of 9.35 GB.

Python was used to process the files in batches. The ETL process consisted of three main stages:

3.1 Extraction

The monthly Parquet files were read using PyArrow. Instead of loading all records into memory at once, the process handled the data in smaller batches.

3.2 Transformation

The data was checked and validated before being loaded into the database.

The validation rules included:

- Checking for missing pickup and drop-off timestamps.
- Checking for invalid trip durations.
- Rejecting records with nonpositive trip times.
- Checking for missing or negative trip distances.
- Checking for missing pickup and drop-off location IDs.
- Identifying unusual fare and distance values using quality flags.

Records that failed the required validation rules were rejected. Other unusual values were flagged for further analysis.

3.3 Loading

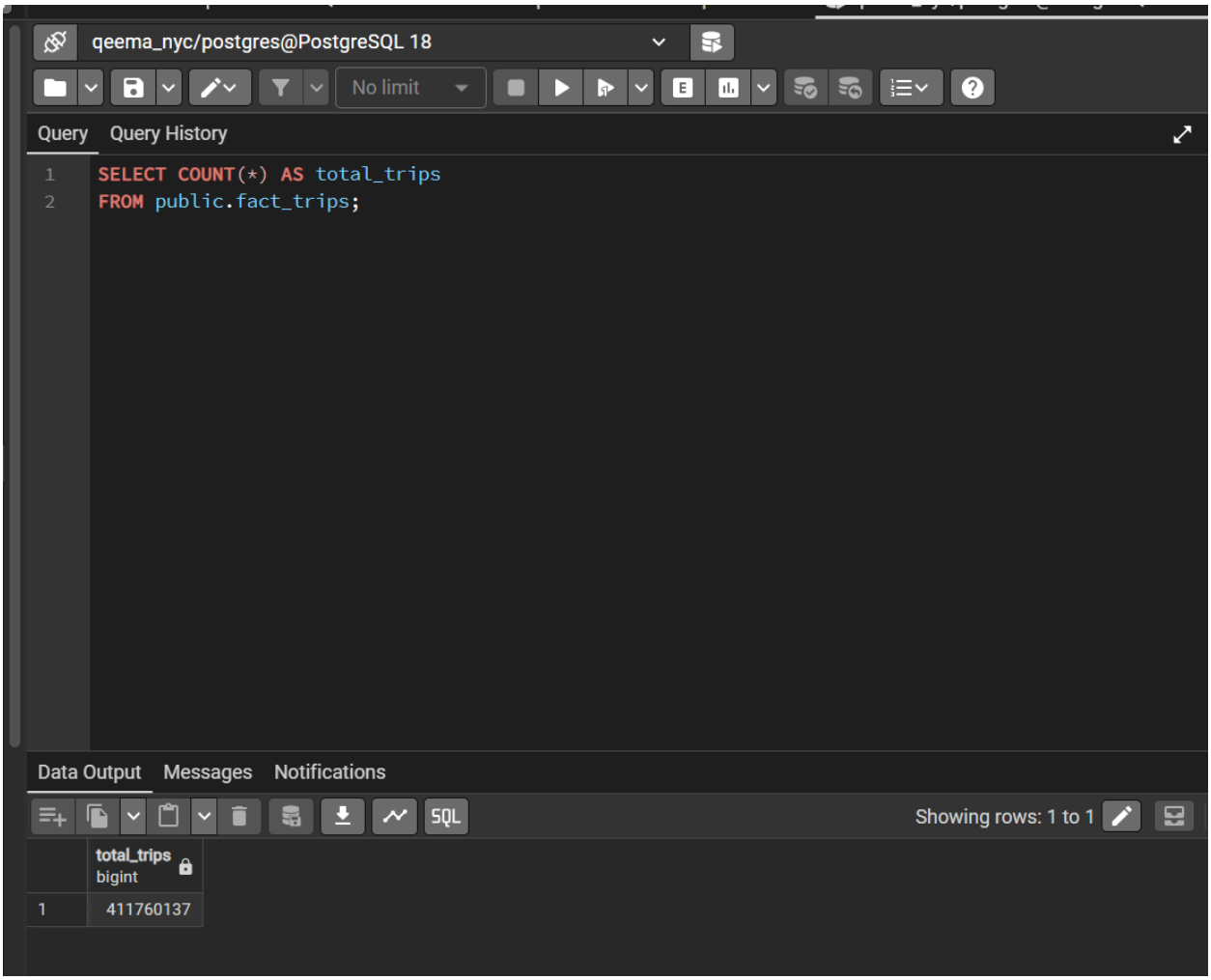
The cleaned records were loaded into PostgreSQL.

The final ETL execution processed all 20 monthly files successfully.

Final ETL results:

Metric	**Result**
Monthly files processed	20
Records loaded into staging	411,760,137
Records loaded into fact table	411,760,137
Rejected records	16,258
Processing status	Completed

```
File    Edit    View
2026-10-31 16:00:51,797 | INFO | Starting staging load: fhvhv_tripdata_2026-01.parquet
2026-10-31 16:28:10,950 | INFO | STAGING COMPLETE: read=20940373 loaded=20939635 rejected=738
2026-10-31 16:38:42,326 | INFO | Dimensions updated for fhvhv_tripdata_2026-01.parquet
2026-10-31 17:11:27,067 | INFO | Fact load completed for fhvhv_tripdata_2026-01.parquet: 20939635 rows
2026-10-31 17:15:13,508 | INFO | COMPLETED fhvhv_tripdata_2026-01.parquet | source=20940373 staging=20939635 fact=20939635 rejected=738
2026-10-31 17:18:45,343 | INFO | Starting staging load: fhvhv_tripdata_2026-02.parquet
2026-10-31 17:44:09,542 | INFO | STAGING COMPLETE: read=19875686 loaded=19875562 rejected=124
2026-10-31 17:55:09,986 | INFO | Dimensions updated for fhvhv_tripdata_2026-02.parquet
2026-10-31 18:25:42,037 | INFO | Fact load completed for fhvhv_tripdata_2026-02.parquet: 19875562 rows
2026-10-31 18:29:51,434 | INFO | COMPLETED fhvhv_tripdata_2026-02.parquet | source=19875686 staging=19875562 fact=19875562 rejected=124
2026-10-31 18:33:11,620 | INFO | Starting staging load: fhvhv_tripdata_2026-03.parquet
2026-10-31 19:01:39,001 | INFO | STAGING COMPLETE: read=22058358 loaded=22057855 rejected=503
2026-10-31 19:13:40,691 | INFO | Dimensions updated for fhvhv_tripdata_2026-03.parquet
2026-10-31 22:30:04,022 | INFO | Fact load completed for fhvhv_tripdata_2026-03.parquet: 22057855 rows
2026-10-31 22:34:28,190 | INFO | COMPLETED fhvhv_tripdata_2026-03.parquet | source=22058358 staging=22057855 fact=22057855 rejected=503
2026-10-31 22:37:59,875 | INFO | Starting staging load: fhvhv_tripdata_2026-04.parquet
2026-10-31 23:05:03,378 | INFO | STAGING COMPLETE: read=20995953 loaded=20995924 rejected=29
2026-10-31 23:17:48,406 | INFO | Dimensions updated for fhvhv_tripdata_2026-04.parquet
2026-11-01 02:38:44,768 | INFO | Fact load completed for fhvhv_tripdata_2026-04.parquet: 20995924 rows
2026-11-01 02:44:07,253 | INFO | COMPLETED fhvhv_tripdata_2026-04.parquet | source=20995953 staging=20995924 fact=20995924 rejected=29
2026-11-01 02:48:34,285 | INFO | Starting staging load: fhvhv_tripdata_2026-05.parquet
2026-11-01 03:17:27,721 | INFO | STAGING COMPLETE: read=22125744 loaded=22125695 rejected=49
2026-11-01 03:30:57,316 | INFO | Dimensions updated for fhvhv_tripdata_2026-05.parquet
2026-11-01 04:06:50,248 | INFO | Fact load completed for fhvhv_tripdata_2026-05.parquet: 22125695 rows
2026-11-01 04:12:03,971 | INFO | COMPLETED fhvhv_tripdata_2026-05.parquet | source=22125744 staging=22125695 fact=22125695 rejected=49
2026-11-01 04:16:30,675 | INFO | Starting staging load: fhvhv_tripdata_2026-06.parquet
2026-11-01 04:43:50,988 | INFO | STAGING COMPLETE: read=20775868 loaded=20773400 rejected=2468
2026-11-01 04:58:32,036 | INFO | Dimensions updated for fhvhv_tripdata_2026-06.parquet
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2026-11-01 08:36:12,352 | INFO | COMPLETED fhvhv_tripdata_2026-06.parquet | source=20775868 staging=20773400 fact=20773400 rejected=2468
2026-11-01 08:40:46,636 | INFO | Starting staging load: fhvhv_tripdata_2026-07.parquet
2026-11-01 09:08:32,024 | INFO | STAGING COMPLETE: read=20921249 loaded=20921218 rejected=31
2026-11-01 09:23:05,956 | INFO | Dimensions updated for fhvhv_tripdata_2026-07.parquet
2026-11-01 09:57:01,498 | INFO | Fact load completed for fhvhv_tripdata_2026-07.parquet: 20921218 rows
2026-11-01 10:02:59,349 | INFO | COMPLETED fhvhv_tripdata_2026-07.parquet | source=20921249 staging=20921218 fact=20921218 rejected=31
2026-11-01 10:08:00,213 | INFO | Starting staging load: fhvhv_tripdata_2026-08.parquet
2026-11-01 10:37:36,140 | INFO | STAGING COMPLETE: read=20493480 loaded=20493410 rejected=70
2026-11-01 10:52:22,229 | INFO | Dimensions updated for fhvhv_tripdata_2026-08.parquet
2026-11-01 14:43:35,589 | INFO | Fact load completed for fhvhv_tripdata_2026-08.parquet: 20493410 rows
2026-11-01 14:49:29,337 | INFO | COMPLETED fhvhv_tripdata_2026-08.parquet | source=20493480 staging=20493410 fact=20493410 rejected=70
```



4. Database Design

PostgreSQL was used to store the processed data and support analytical queries.

A star schema was implemented to organize the data and make analysis easier.

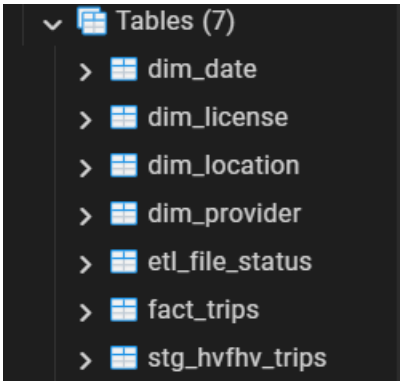
The main fact table is fact_trips, which stores the trip records and connects to the dimension tables.

The dimension tables are:

- dim_date: Stores date-related information.
- dim_location: Stores pickup and drop-off location information.
- dim_license: Stores license-related information.
- dim_provider: Stores provider information.

The fact table contains the main trip measures, including trip distance, trip duration, fares, tips, and other trip-related values.

This structure makes it easier to analyze trips across different dates, locations, and providers.



5. SQL Analysis

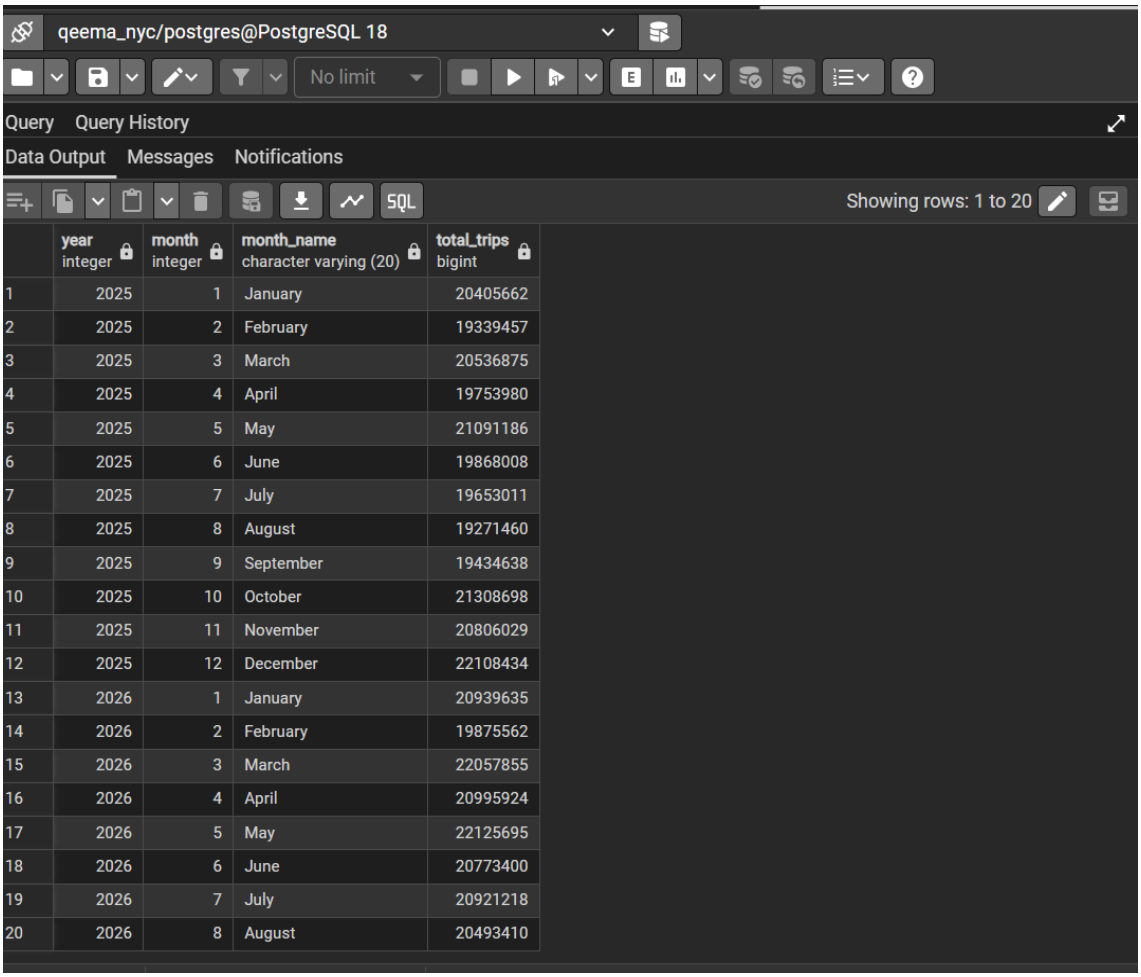
SQL was used to analyze the data and answer important business questions.

The analysis focused on monthly trip activity, pickup locations, borough performance, trip duration, providers, and tipping behavior.

5.1 Monthly Trip Volume

The monthly analysis was used to understand how trip demand changed over time.

A monthly summary was created to calculate the total number of trips for each month.



	year integer	month integer	month_name character varying (20)	total_trips bigint
1	2025	1	January	20405662
2	2025	2	February	19339457
3	2025	3	March	20536875
4	2025	4	April	19753980
5	2025	5	May	21091186
6	2025	6	June	19868008
7	2025	7	July	19653011
8	2025	8	August	19271460
9	2025	9	September	19434638
10	2025	10	October	21308698
11	2025	11	November	20806029
12	2025	12	December	22108434
13	2026	1	January	20939635
14	2026	2	February	19875562
15	2026	3	March	22057855
16	2026	4	April	20995924
17	2026	5	May	22125695
18	2026	6	June	20773400
19	2026	7	July	20921218
20	2026	8	August	20493410

5.2 Pickup Location Analysis

The top 10 pickup locations were identified based on the total number of trips.

The results showed that LaGuardia Airport and JFK Airport were the two busiest pickup locations in the dataset.

5.3 Average Fare by Borough

The average base passenger fare was calculated for each borough.

Manhattan had the highest average fare among the major boroughs, at approximately $32.20, while the Bronx had the lowest, at approximately $19.71.

5.4 Trips by Day of the Week

The analysis showed that Saturday had the highest trip volume, with approximately 68.2 million trips. Monday had the lowest, with approximately 51.0 million trips.

This suggests that trip demand varies depending on the day of the week.

5.5 Trip Duration by Borough

The average trip duration was compared across boroughs.

Manhattan had the longest average trip duration among the major boroughs, at approximately 21.11 minutes.

Staten Island had the shortest, at approximately 15.84 minutes.

5.6 Trips by Provider

The provider analysis showed that two provider bases accounted for almost all trips in the dataset.

Provider B03404 represented approximately 72.21% of trips, while B03406 represented approximately 27.79%.

5.7 Trips by Hour

The hourly analysis showed that trip activity was highest around 6 PM, with approximately 24.1 million trips.

The lowest activity occurred around 3 AM.

5.8 Tip Analysis

Manhattan had the highest tipping frequency among the major boroughs, with approximately 24.47% of trips receiving tips.

The Bronx had the lowest tipping frequency, at approximately 7.90%.

These results provide a better understanding of passenger behavior and trip activity across different areas and times.

6. Performance Optimization

Performance optimization was important because the fact table contains more than 411 million records.

Two optimization techniques were implemented.

6.1 Materialized View

A materialized view called monthly_trip_summary was created to store precomputed monthly trip totals.

Before using the materialized view, the original aggregation query took approximately 142.90 seconds.

Reading the already calculated monthly summary took approximately 0.040 milliseconds in the measured test.

This demonstrates the benefit of storing frequently used aggregate results instead of recalculating them from the entire fact table every time.

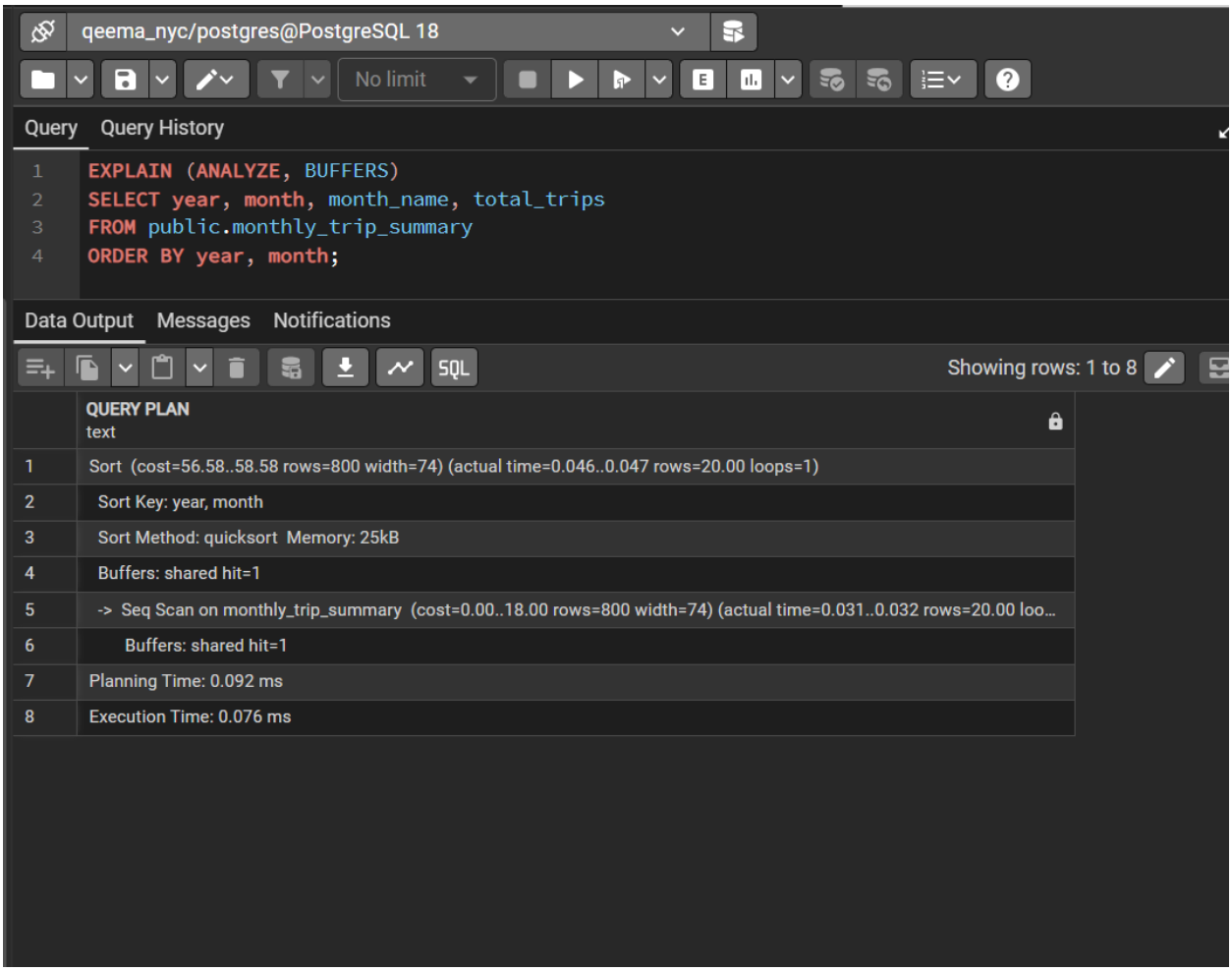
The materialized view must be refreshed when the underlying data changes.

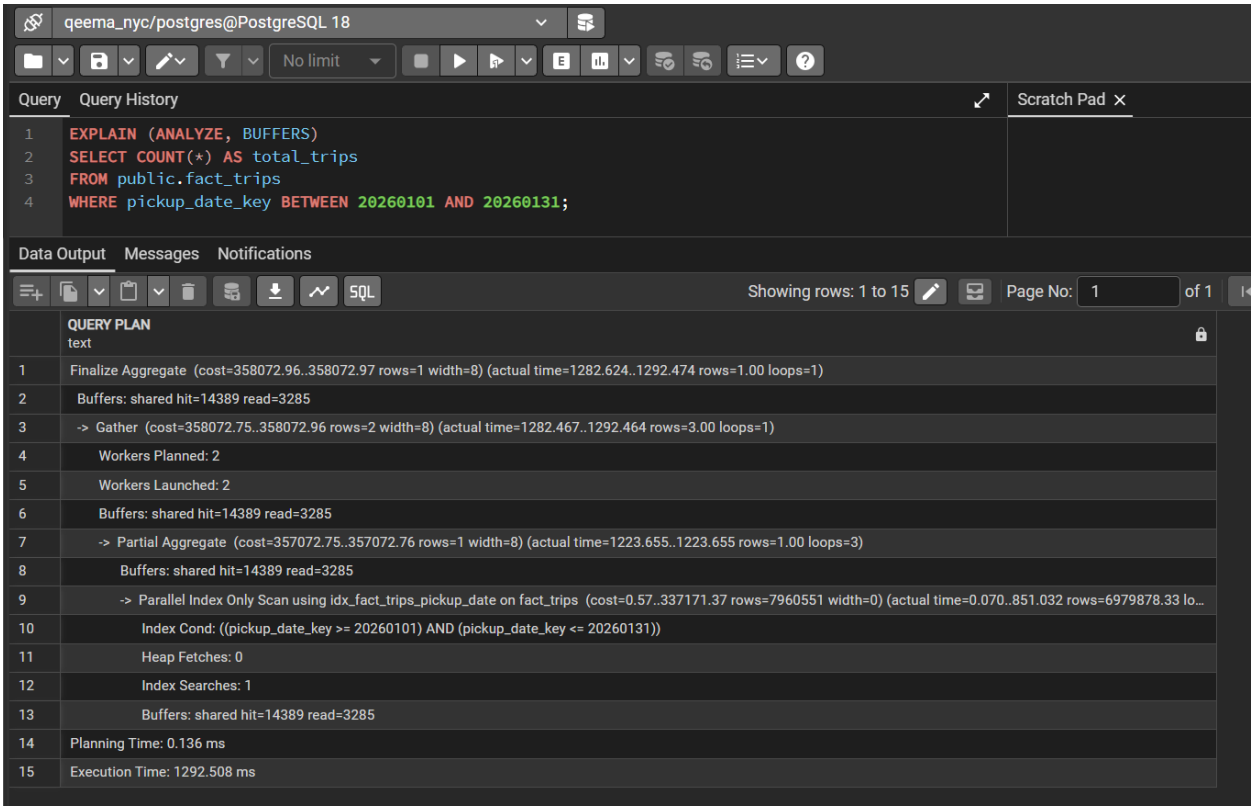
6.2 Database Indexing

An index called idx_fact_trips_pickup_date was created on the pickup_date_key column in the fact table.

The index was tested using a query that counted trips during January 2026.

The execution plan confirmed that PostgreSQL used a parallel index-only scan, with zero heap fetches. The query completed in approximately 2.07 seconds.





7. Apache Airflow

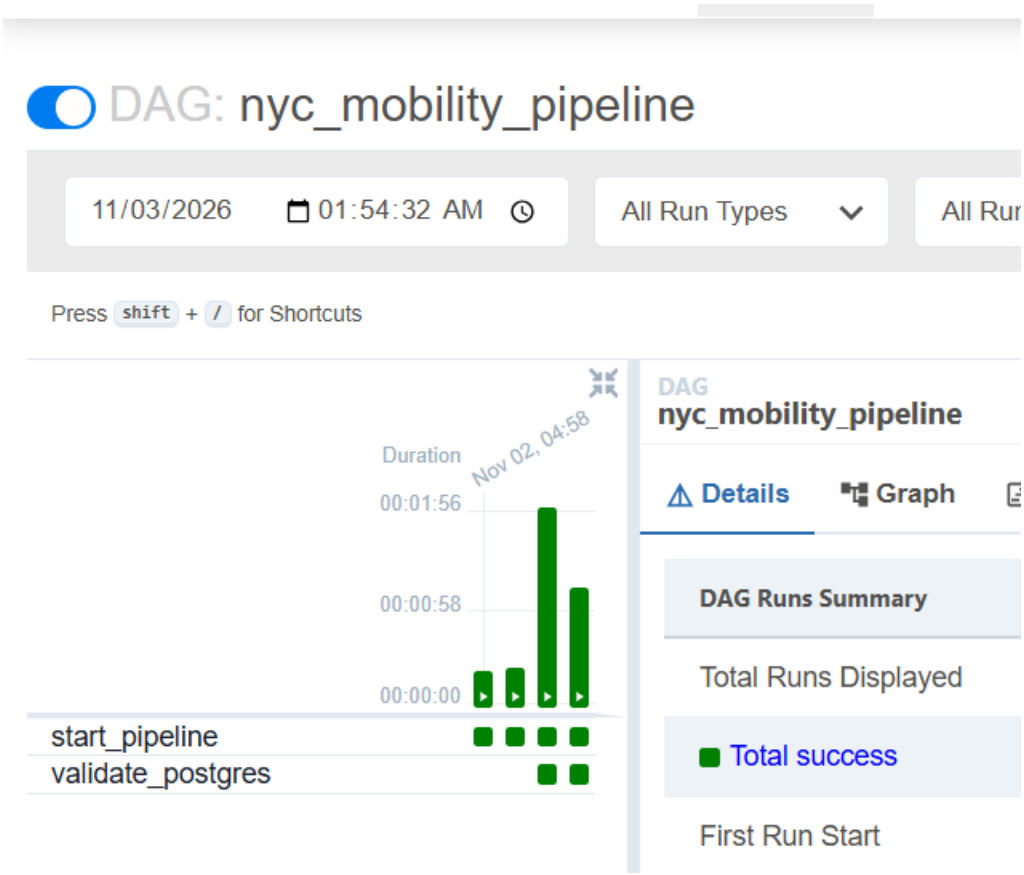
Apache Airflow was configured to monitor and validate the PostgreSQL database.

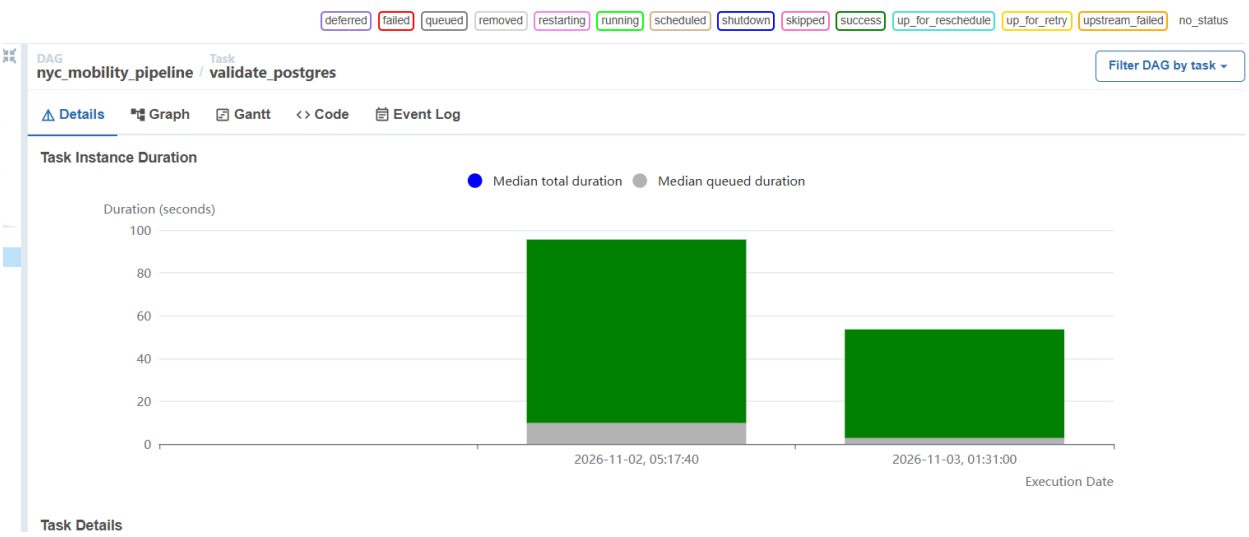
A DAG called nyc_mobility_pipeline was created with the following tasks:

- Start the workflow.
- Connect to PostgreSQL.
- Validate the database by checking the number of records in the fact_trips table.

The DAG executed successfully and confirmed that the fact table contained 411,760,137 records.

The Airflow workflow was used for database validation and monitoring. The main ETL loading process was executed separately using Python.





8. Dashboard Development

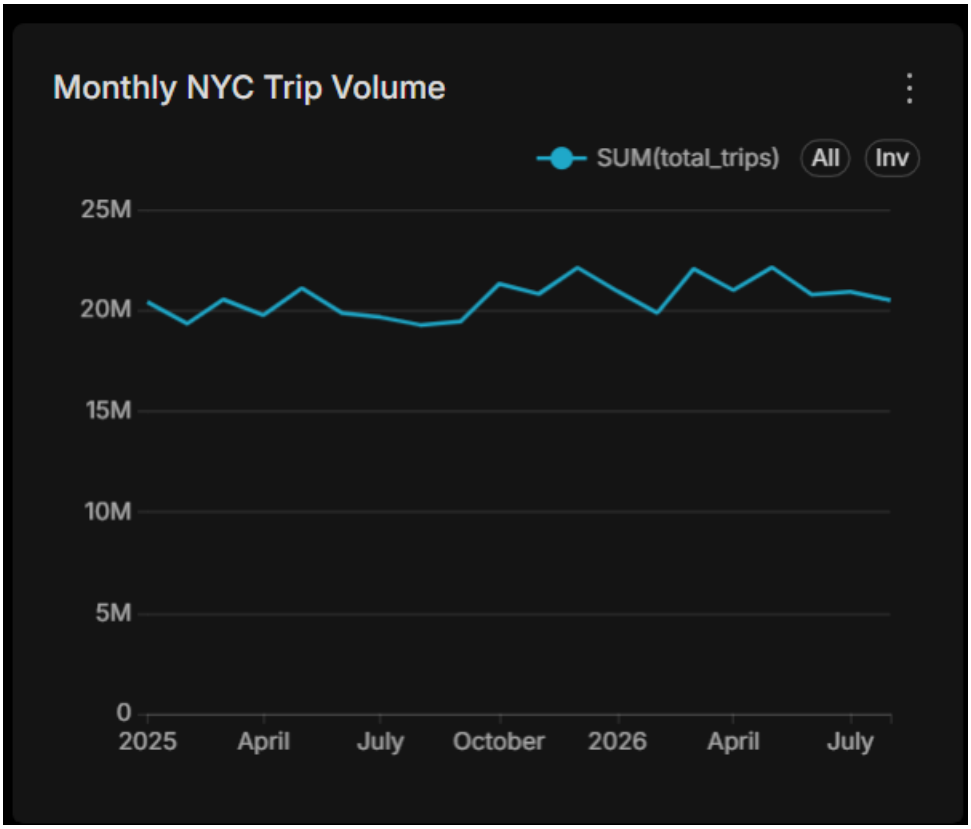
Apache Superset was used to create an interactive dashboard called **NYC Mobility Analytics Dashboard**.

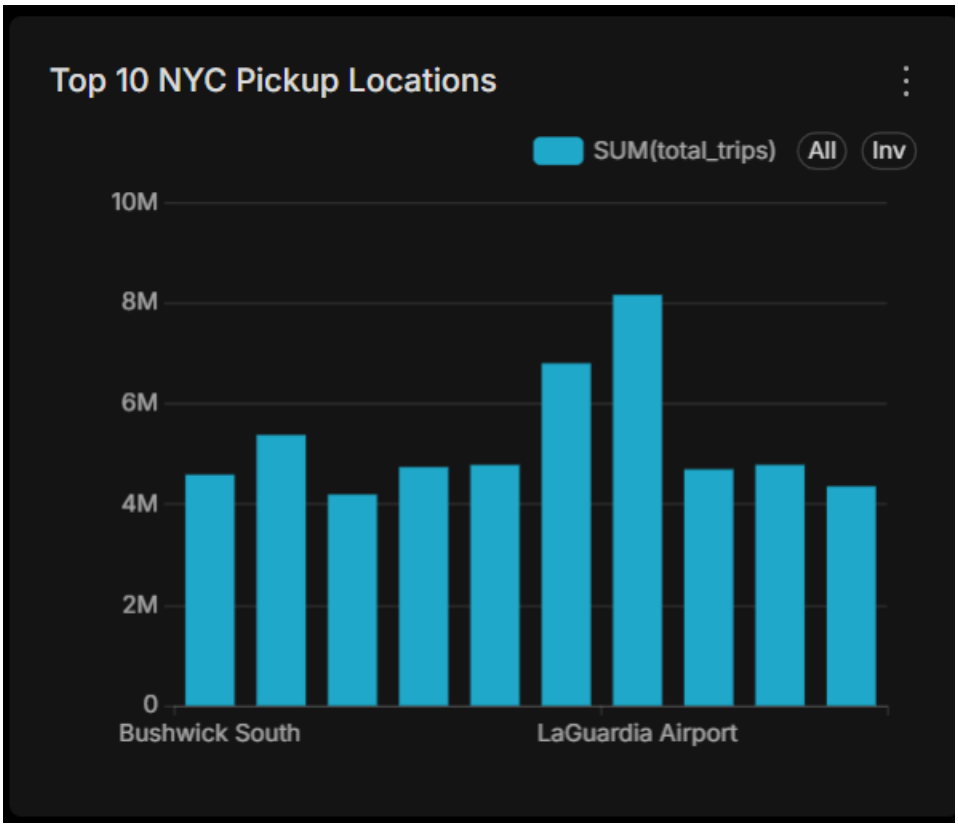
The dashboard connects to the PostgreSQL database and presents the analysis through eight visualizations:

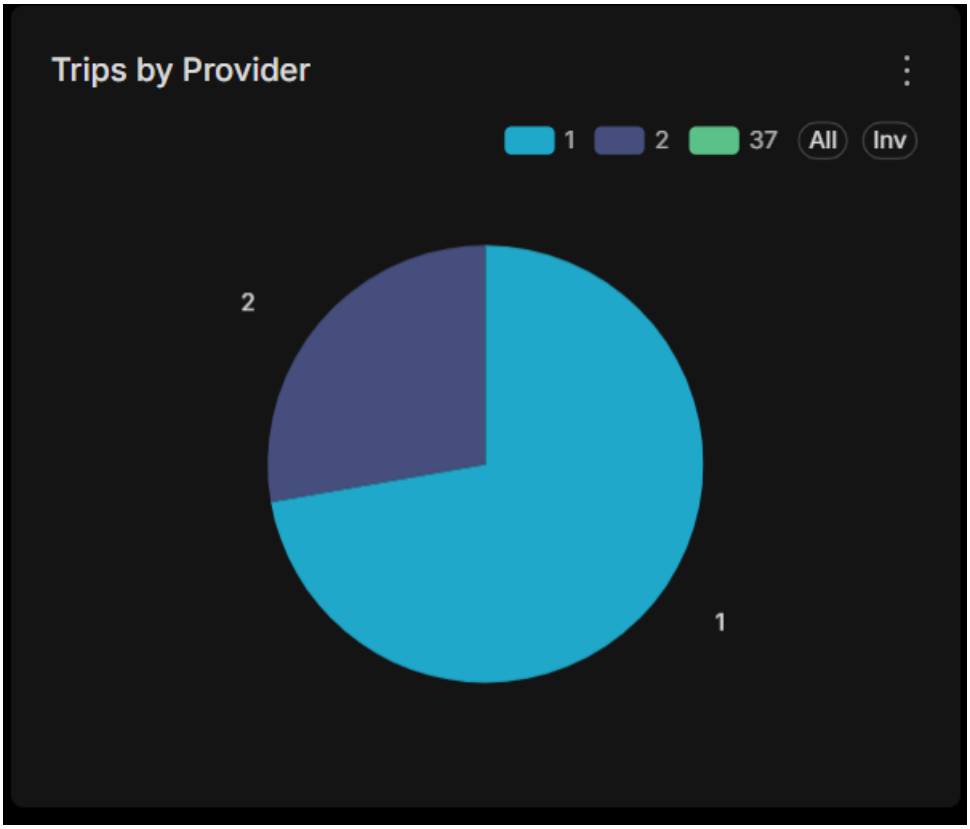
1. **Monthly NYC Trip Volume:** Shows changes in trip demand over time.
2. **Top 10 NYC Pickup Locations:** Identifies the busiest pickup locations.
3. **Average Fare by Borough:** Compares average base passenger fares across boroughs.
4. **Trips by Day of Week:** Shows how trip volume changes throughout the week.
5. **Average Trip Duration by Borough:** Compares trip duration across boroughs.
6. **Trips by Provider:** Shows the distribution of trips between providers.
7. **Trips by Hour:** Identifies the busiest and quietest hours.
8. **Tip Analysis by Borough:** Compares tipping frequency and average tips across boroughs.

The dashboard provides a visual overview of the dataset and makes the results easier to understand.




Monthly NYC Trip Volume
SUM(total_trips)
All
Inv
25M
20M
15M
10M
5M
0
2025
April
July
October
2026
April
July


Top 10 NYC Pickup Locations
SUM(total_trips)
All
Inv
10M
8M
6M
4M
2M
0
Bushwick South
LaGuardia Airport



9. Conclusion

This project provided practical experience in building a data analytics platform using a large real-world dataset.

The work covered the main stages of a data engineering and analytics workflow, including data extraction, transformation, validation, database loading, dimensional modeling, SQL analysis, performance optimization, workflow validation, and dashboard development.

The final database contains more than 411 million processed trip records, and the Superset dashboard provides several views of NYC mobility patterns.

Overall, the project helped me understand how different data engineering and business intelligence tools work together to transform raw data into useful information for analysis and decision-making.